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LinkedIn | GitHub | Website

Backend developer focused on building reliable systems, monitoring tools, and data pipelines that operate in real-world environments. For a more accurate and up-to-date overview, please visit my **website**, as well as my **GitHub** and **LinkedIn** profiles.

ShielD – Real-Time Anomaly Detection for System Logs

ShielD is a lightweight Java utility that monitors system logs and web traffic integrity in real time. It ingests log or event streams, applies rule-based detection for suspicious patterns (e.g., repeated failed logins or request bursts), and can optionally use TensorFlow Java / DJL anomaly scoring for higher-signal detection. The system emits real-time console alerts and structured JSON reports for automation and CI/CD integration. Configurable thresholds and modular detectors allow sensitivity-tuning without code changes, keeping operational overhead low for small servers and development environments.

Stack: Java, Maven, TensorFlow Java / DJL (optional), Tablesaw
ShielD Github Repository

InfrawatchLite – Lightweight Linux Performance Monitor

InfrawatchLite is a portable Python monitoring utility designed to detect resource spikes and support rapid system diagnosis. It collects CPU, memory, disk, network, and optional GPU metrics via psutil, with thresholds and sampling defined through JSON/YAML. When limits are exceeded, it produces console alerts, structured JSON/CSV logs, and top-process attribution to accelerate root-cause analysis. Modular collectors, event-driven alerts, and cooldown timers prevent alert fatigue while keeping the tool lightweight; optional visualizations support offline performance analysis.

Tools: Python, PyTorch, psutil, JSON, Bash, Git
InfrawatchLite GitHub Repository

HelixNet – Machine Learning for Protein Structure Prediction

HelixNet is a machine learning pipeline that classifies protein sequences into Helix, Sheet, or Coil with ~97% accuracy. Built in Java using Tensor processing libraries, it ingests FASTA/CSV sequence data, applies one-hot or n-gram embeddings, and trains reproducible models with clear separation of training, validation, and test datasets. The system outputs predictions alongside detailed evaluation of metrics (accuracy, precision/recall, F1 scores, confusion matrices) and visual training curves to support model analysis. Modular data loaders and model builders enable rapid architecture experimentation, while deterministic seeds and saved artifacts ensure reproducibility, making HelixNet suitable for research workflows and production-style pipelines.

Tools: Java, TensorFlow Java / DL4J, Git, Bash
HelixNet GitHub repository